

Chapter 47: Fetal-Growth Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2141–2181 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 7 figures are included from the PDF.

Big picture: Growth disorders sit at both ends — **fetal-growth restriction (FGR)** and **macrosomia**. Most small or large babies are simply **constitutionally** small or large. FGR is managed with **serial growth scans, umbilical artery Doppler and fetal testing**, with delivery timed by the Doppler findings. For suspected macrosomia, **neither early induction nor elective cesarean** is advised below set weight thresholds.

1. Normal Fetal Growth

Phase	Weeks	Process	Growth rate
1	0–16	Hyperplasia (↑ cell number)	~ 5 g/d at 15 wk
2	16–32	Hyperplasia + hypertrophy	~ 15 g/d at 24 wk
3	>32	Hypertrophy ; most fat and glycogen laid down	~ 30 g/d at 34 wk

- NICHD Fetal Growth Studies (1733 low-risk women): growth velocity **peaks at 35 wk**.
- Growth varies widely and correlates poorly with birthweight percentile: some fetuses <5th percentile keep their velocity and end up heavier than initially bigger fetuses → **serial sonography** matters.

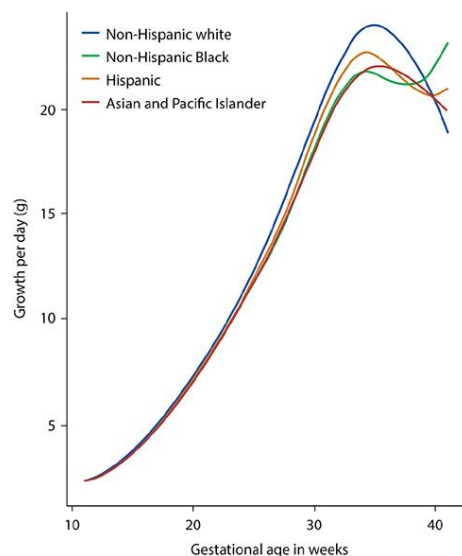


Figure 47-1. Fetal weight gain (g/day) by race/ethnicity: velocity peaks ~35 weeks at ~22–24 g/day, then diverges.

Normal birthweight

- Accurate GA (obstetrical estimate incl. sonography) is critical.
- Percentiles below: **>3 million** US singletons, 2011 (Duryea 2014).
- LMP-based dating gives **heavier weights for GA**, especially preterm → affects SGA/LGA prevalence.
- Duryea curve = **population reference** (all risks and outcomes), not a **standard** (normal pregnancies, normal outcomes). Because it includes preterm births, it may overestimate impaired growth.

Table 47-1. 2011 Gestational Age Birthweight (g) Percentiles for 3,252,011 Singleton Live Births in the United States

Age (wk)	5th	10th	50th	90th	95th
24	539	567	680	850	988
25	540	584	765	938	997
26	580	637	872	1080	1180
27	650	719	997	1260	1467
28	740	822	1138	1462	1787
29	841	939	1290	1672	2070
30	952	1068	1455	1883	2294
31	1080	1214	1635	2101	2483
32	1232	1380	1833	2331	2664
33	1414	1573	2053	2579	2861
34	1632	1793	2296	2846	3093
35	1871	2030	2549	3119	3345
36	2117	2270	2797	3380	3594
37	2353	2500	3025	3612	3818
38	2564	2706	3219	3799	3995
39	2737	2877	3374	3941	4125
40	2863	3005	3499	4057	4232
41	2934	3082	3600	4167	4340
42	2941	3099	3686	4290	4474

From Duryea, 2014, with permission.



Figure 47-2. LMP-based dating overstates birthweight for GA at 28–36 weeks compared with the obstetrical (ultrasound-informed) estimate.

Physiology of fetal growth

- **Substrate supply and placental transfer** drive development; **the genome** sets growth potential.
- **Insulin and insulin-like growth factors** are key regulators. **Leptin** rises with gestation and correlates with birthweight and fat mass.

- **Glucose:** FGR neonates usually do **not** have low cord glucose; deprivation restricts growth only with **long-term severe caloric deprivation**. **Hyperglycemia** more consistently causes overgrowth.
 - **HAP0:** ↑ **cord C-peptide** (fetal hyperinsulinemia) → ↑ birthweight, even with maternal glucose below diabetic thresholds.
- **Lipids:** maternal free fatty acids (late pregnancy) and **omega-3** intake correlate with birthweight.
 - **Endothelial lipase** ↓ in FGR placentas, **overexpressed in diabetes**; obesity ↑ trophoblast fatty-acid transport proteins → lipid accumulation, placental inflammation.

2. Fetal-Growth Restriction: Definition

- An EFW threshold identifies an **at-risk category**, not disease. A 10th-percentile cutoff labels ~10% of low-risk fetuses (mostly constitutionally small) and misses fetuses above it that have not reached their potential.
- **ACOG/SMFM definition of FGR: EFW <10th percentile OR abdominal circumference (AC) <10th percentile** for GA.
 - Adding AC finds **1 extra SGA newborn per 14 sonograms** in the last month.
 - Management is further stratified by lower thresholds (**<3rd percentile**), **oligohydramnios** and **umbilical artery Doppler**.
- **Customized growth charts** (maternal height/weight, ethnicity, parity, fetal sex) **do not improve outcomes** → **not recommended**.

FGR	SGA
Prenatal sonographic diagnosis	Postnatal designation — birthweight <10th percentile

- **50%** of fetuses with suspected FGR on routine third-trimester scans had birthweight >10th percentile; false positives were delivered **2 weeks earlier** (iatrogenic harm).
- **Up to 70% of SGA newborns are not pathologically growth restricted** (normal when maternal ethnicity, parity, weight, height are considered).
- Maternal and paternal birthweights explain **6% and 3%** of birthweight variance.
- **Most adverse outcomes occur below the 3rd percentile.** Parkland (>80,000 term births): neonatal mortality and most morbidity rose only at **≤3rd percentile**.
- Small-but-normal neonates stay smaller to 2 yr but show **no metabolic derangement**.

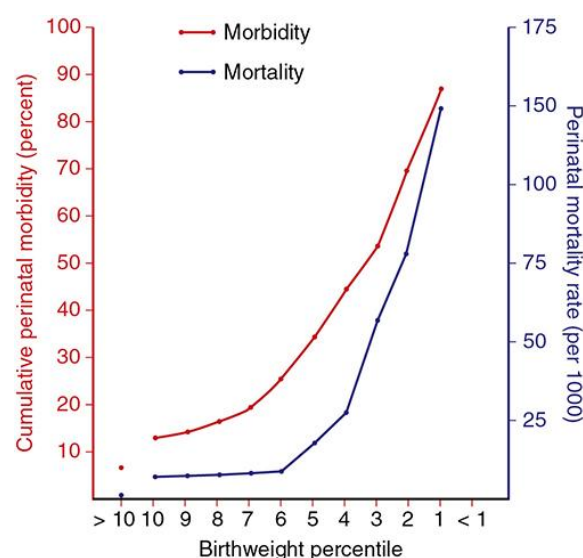


Figure 47-3. In 1560 SGA fetuses, morbidity and perinatal mortality rise steeply below the 3rd–4th percentile (mortality ~150/1000 at the 1st percentile).

3. Detection

Three tenets:

1. **Accurate GA**, ideally by first- or early second-trimester sonography. If GA uncertain and FGR suspected → serial sonography.
2. **Low-risk**: suspect when **fundal height lags by ≥ 3 cm after 24 wk** → sonography.
3. **At-risk pregnancy**: consider sonography at **~32 wk**; sometimes **every 4 weeks** (e.g., **twins**, where fundal height cannot assess each fetus).

Fundal height

- Measured each visit **after 20 wk**; cm $\approx$ weeks. Discrepancy **≥ 3 cm** without explanation → ultrasound.
- **The only routine FGR screening endorsed by ACOG/SMFM**; simple, safe, cheap.
- **Sensitivity only 11–25%**; worse in **obesity**.

Sonography

- Routine third-trimester scans detect **<20% to >60%** of SGA newborns — misses late-onset FGR; EFW is imperfect.
- FGR **recurrence $\approx 20\%$** (often small mothers with small healthy babies).
- **Routine third-trimester ultrasound is NOT recommended** — no maternal or fetal benefit (Cochrane: 13 trials, 34,980 women).

4. Pathophysiology

- FGR is one of the "**major obstetrical syndromes**" of **defective early placentation** (with preeclampsia) — vascular and immunological mechanisms.
- **Corin** (atrial natriuretic peptide–converting enzyme) is needed for trophoblast invasion and spiral artery remodeling; corin-deficient mice get FGR-like and preeclampsia features; corin mutations occur in preeclampsia.
- **Immunological**: maternal rejection of the "paternal semiallograft".
 - **C4d** (complement, humoral rejection) linked to **chronic villitis — 88% vs 5%** of controls — and small placentas.
 - Chronic villitis → placental hypoperfusion, fetal acidemia, FGR.

5. Risk Factors

Three overlapping compartments: **mother, fetus, placenta**.

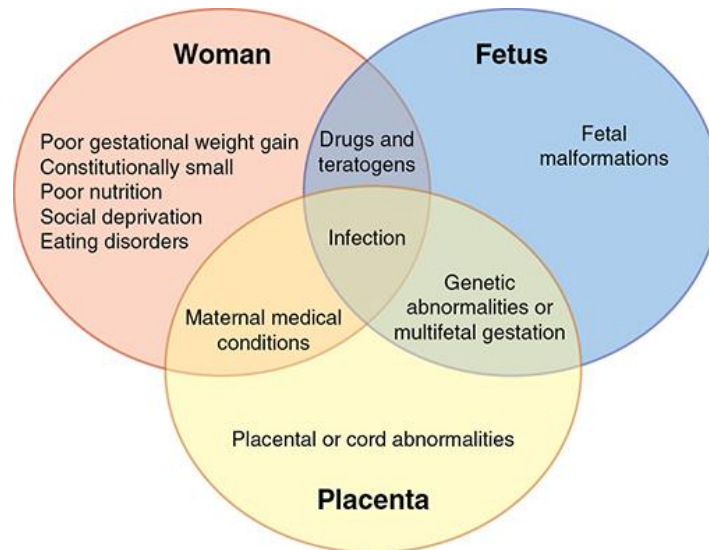


Figure 47-4. FGR risk factors overlap across woman (nutrition, weight gain, social deprivation), fetus (malformations) and placenta (placental/cord abnormalities); infection sits in all three.

Maternal factors

Factor	Key facts
Gestational weight gain / nutrition	Weight gain correlates with fetal size. Second/third-trimester gain below IOM advice → ↑ SGA in all BMI groups except class II–III obesity . Dutch famine 1944 (500 kcal/d for 6 months) → mean birthweight ↓ 250 g . Micronutrients may ↓ LBW (Indonesia; Cochrane 141,849 women).
Socioeconomic	Linked to smoking, alcohol/drugs, poor nutrition, food insecurity, late prenatal care. Interventions ↓ LBW. Racial gap persists even with equal access (military).
Vascular and renal disease	Chronic hypertension (esp. + superimposed preeclampsia); abnormal early uterine artery Doppler → preeclampsia, SGA, delivery <34 wk. Ischemic heart disease → 16% SGA . Nephropathy.
Pregestational diabetes	FGR from malformations or vasculopathy; worsens with White class , esp. nephropathy
Chronic hypoxia	Asthma, cyanotic heart disease, lung disease, smoking, altitude. Smoking: dose-dependent, ~200 g lighter . Altitude: –150 g per 1000 m .
Anemia	Usually no effect , except sickle-cell and other inherited anemias. Poor plasma-volume expansion is linked to FGR.
Antiphospholipid syndrome	Anticardiolipin, lupus anticoagulant, anti-β2 glycoprotein-I (strongest link, esp. early-onset). " Two-hit ": endothelial damage → intervillous thrombosis. >1 antibody type → higher risk.
Drugs and substances	Cyclophosphamide/antineoplastics, lead , cocaine, methamphetamine; alcohol (FGR is a criterion for fetal alcohol syndrome); smoking → 2–3× FGR . Most drugs do not affect growth.

Placental, cord and uterine

- Chronic **abruption**, extensive **infarction**, **chorioangioma**, **velamentous cord insertion**, **umbilical artery thrombosis**, abnormal implantation (preeclampsia), some **uterine malformations**.
- **Confined placental mosaicism** is a recognized cause.

Multifetal gestation

- Growth of one or more fetuses is more often impaired → **serial sonography**. Parkland uses a **chorionicity-specific twin nomogram**; also assess **discordance** (Ch 48).

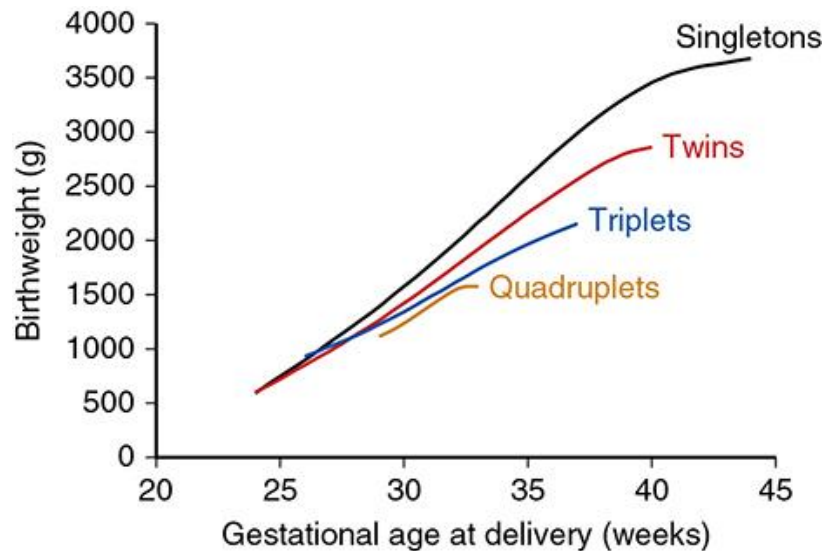


Figure 47-5. Mean birthweight by GA falls progressively from singletons to twins, triplets and quadruplets, diverging after ~30 weeks.

Infections (up to 5% of FGR)

- **Rubella and CMV** best known (fetal calcifications from cell death; earlier infection = worse).
 - Congenital rubella (Vietnam): **39%** LBW.
 - CMV: no severe cases if primary infection **after 14 wk**; FGR in **30%** of those with sonographic findings.
- **Tuberculosis**: OR **1.7** for LBW.
- **Syphilis**: placenta is paradoxically **larger and heavier** (edema, perivascular inflammation); strongly linked to **preterm birth**.
- **Toxoplasmosis**: LBW in ~**40%** (only 13% treated antenatally). **Malaria** → LBW; give prophylaxis early.

Congenital malformations

- Major malformations/chromosomal abnormalities → FGR at **2×** the population rate.
- **Gastroschisis = the defect most strongly linked to FGR**: $\frac{1}{3}$ <10th percentile.
- Isolated cardiac defects: SGA **13%** (3% above baseline).
- FGR + structural anomaly → ↑ risk of a genetic syndrome → **offer amniocentesis with chromosomal microarray**.

Genetic abnormalities

Condition	SGA prevalence / feature
Trisomy 21	15–30%
Trisomy 13	~50%; short CRL
Trisomy 18	>80%; FGR + anomalies + hydramnios ; short CRL
Turner syndrome	Severity $\propto$ haploinsufficiency of Xp
Extra X chromosomes	Poor growth not characteristic

6. Management of FGR

- Stillbirth risk ↑; **early-onset** FGR is especially problematic.
- **General tenets**:
 - **Growth scans every 3 weeks**
 - **At least weekly amniotic fluid + umbilical artery (UA) Doppler**
 - **Fetal well-being testing** (NST or BPP)

- Slow-but-progressing EFW is more reassuring than a **plateau**
- Consider a **detailed anatomic survey** and **amniocentesis** (genetics, infection), especially with early onset

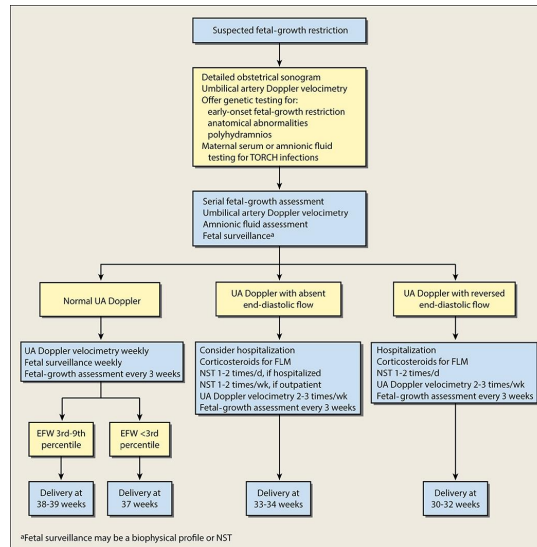


Figure 47-6. FGR algorithm: delivery timing is set by UA Doppler and EFW — normal Doppler: 38–39 wk (EFW 3rd–9th %ile) or 37 wk (<3rd %ile); AEDF: 33–34 wk; REDF: 30–32 wk.

Finding	Surveillance	Deliver at
Normal UA Doppler, EFW 3rd–9th %ile	Weekly UA Doppler and fetal testing; growth every 3 wk	38–39 wk
Normal UA Doppler, EFW <3rd %ile	Same	37 wk
Absent end-diastolic flow (AEDF)	Consider admission; steroids ; NST 1–2×/day inpatient (1–2×/wk outpatient); UA Doppler 2–3×/wk; growth every 3 wk	33–34 wk
Reversed end-diastolic flow (REDF)	Admit; steroids ; NST 1–2×/day; UA Doppler 2–3×/wk; growth every 3 wk	30–32 wk
Oligohydramnios (near term)	—	36 ⁰ / ₇ –37 ⁶ / ₇ wk

Memory hook: "Normal 37–39, Absent 33–34, Reversed 30–32" — each step of Doppler deterioration moves delivery earlier.

Delivery timing trials

- **GRIT** (Thornton 2004; 548 women, **24–36 wk**): immediate vs delayed delivery → **no difference** in death or disability at 2 yr, nor at 6–13 yr.
- **DIGITAT** (321 women, **≥36 wk**): induction vs expectant → **no difference** in composite neonatal morbidity (fewer admissions if induced after 38 wk, secondary analysis); same 2-yr neurodevelopment.

Doppler velocimetry

- **UA Doppler is central** to FGR management. Progression: ↑ **impedance** (↑ **S/D**) → **absent EDF** → **reversed EDF**, paralleling hypoxia, acidosis and death.
- EFW <10th percentile: adverse outcome **1%** with normal UA Doppler vs **12%** with abnormal (O'Dwyer 2014, 1116 fetuses).
- **Stillbirth risk: AEDF 7%, REDF 19%.**
- **ACOG/SMFM: serial UA Doppler** in FGR.

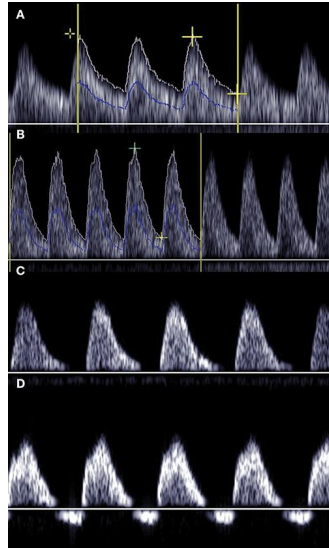


Figure 47-7. Umbilical artery waveforms: (A) normal S/D, (B) raised S/D, (C) absent end-diastolic flow, (D) reversed end-diastolic flow.

Other vessels — NOT recommended for routine management

- **Ductus venosus:** abnormal flow = $\uparrow$ central venous pressure ($\downarrow$ cardiac compliance, $\uparrow$ RV end-diastolic pressure) $\rightarrow$ stillbirth **20%, 46% with reversed a-wave**.
- **Umbilical vein pulsatility** $\rightarrow$ cardiac dysfunction.
- **Cerebroplacental ratio (CPR) = MCA PI $\div$ UA PI;** reflects **brain-sparing** vasodilation. **CPR <1** $\rightarrow$ earlier delivery, lower birthweight, cesarean, NICU, death — but a metaanalysis (18,731 fetuses) found it **not predictive**.

Near-term fetus

- Normal UA Doppler, normal fluid, reassuring testing $\rightarrow$ delivery can wait until **37–38 wk**.
- **Oligohydramnios** $\rightarrow$ **deliver 36⁰/₇–37⁶/₇ wk**.
- Normal FHR $\rightarrow$ plan vaginal delivery; some fetuses **do not tolerate labor**.

Fetus remote from term (<34 wk)

- Normal fluid and testing $\rightarrow$ **observe**; reassess growth **no sooner than 3 weeks**; weekly outpatient UA Doppler, fluid and testing.
- **AEDF or REDF** $\rightarrow$ **inpatient surveillance**.
- **No treatment works:** bed rest, nutrient supplements, plasma volume expansion, **oxygen**, antihypertensives, **heparin, aspirin** — all ineffective.
- Balance fetal death risk vs prematurity. 2-yr neurodevelopment is best predicted by **birthweight and GA**; Doppler abnormalities do not reliably predict childhood cognition.

Intrapartum management

- Placental insufficiency may worsen in labor; **oligohydramnios** $\rightarrow$ **cord compression** $\rightarrow$ $\uparrow$ **cesarean**, neonatal hypoxia, **meconium aspiration**.
- Skilled neonatal resuscitation at delivery.
- Severely growth-restricted newborn: **hypothermia, hypoglycemia, polycythemia, hyperviscosity** — worst at lowest weights.

7. Outcomes of SGA/FGR

- Battaglia and Lubchenco (1967): SGA = **<10 th percentile**; mortality at 38 wk **1% vs 0.2%**.
- SGA stillbirth rate **$\sim 1\%$ ($2\times$ the population)**.
- **<27 wk and <10 th percentile: $\sim 4\times$ death or neurodevelopmental impairment; $\sim 3\times$ cerebral palsy.**

- **<5th percentile** (91,000 pregnancies): ↑ low Apgar, RDS, NEC, sepsis; stillbirth **6×**, neonatal death **4×**.
- **<1st percentile**: only **14%** survived to discharge.
- Poor motor, cognitive, language, attention and behavioral outcomes persist into adolescence.

Early-onset FGR

- **~30%** of FGR is diagnosed **before 32 wk** — the cutoff for **early vs late onset** (SMFM).
- Early onset: more **severe**, more **placental dysfunction and maternal hypertension**, more **UA Doppler abnormalities**, ↑ antiangiogenic **sVEGFR-1 / sFlt-1** (as in preeclampsia).

	Early-onset FGR	Late-onset FGR
Timing	<32 wk (~30%)	≥32 wk
Severity	More severe	Milder
Association	Placental dysfunction, hypertension/preeclampsia , ↑ sFlt-1	Less placental disease
UA Doppler	Often abnormal	Less often abnormal

Barker hypothesis

- Fetal and infant health (under- and overgrowth) programs adult disease: **hypertension, atherosclerosis, type 2 diabetes, metabolic syndrome**. Early postnatal weight gain also matters.
- **Heart**: SGA infants (<34 wk) have a more **globular ventricle** with systolic and diastolic dysfunction; LBW adolescents have a thicker LV posterior wall.
- **Kidney**: low or high birthweight, maternal obesity and GDM harm renal development.

Prevention

- Preconception: treat maternal disease, adjust medications, **stop smoking**, antimalarial prophylaxis, correct nutrition.
- **Treating mild-moderate hypertension does not ↓ SGA.**
- **No drug prevents FGR.** Aspirin: no benefit in a 1700-woman RCT; metaanalyses showed ↓ FGR, but in women at risk for preeclampsia (secondary finding) → **aspirin NOT recommended for FGR prevention** without preeclampsia risk factors.

8. Fetal Macrosomia

Definition

- **Macrosomia**: EFW above a threshold — **4000, 4500 or 5000 g** (no consensus). <4000 g is not excessively large.
- **LGA**: weight **>90th percentile** for GA; usually constitutional. The 90th percentile at **39 wk ≈ 3900 g** → most LGA infants are not macrosomic.
- US 2019: **4000–4499 g 6.4%**; **4500–4999 g 0.9%**; **≥5000 g 0.1%**.
- Parkland (>350,000 births): only **1.4% ≥4500 g**.
- Williams: abnormal overgrowth = **>2 SD above the mean** (~3% of births) ≈ **4500 g at 40 wk**.

Table 47-2. Birthweight Distribution of 354,509 Liveborn Infants at Parkland Hospital between 1988 and 2012

Birthweight (g)	Births No.	Births %	Maternal Diabetes No.	Maternal Diabetes %
500–3999	322,074	90.9	13,365	86.2
4000–4249	19,106	5.4	1043	6.7
4250–4499	8391	2.4	573	3.7
4500–4749*	3221	0.9	284	1.8
4750–4999	1146	0.3	134	0.7
5000–5249	385	0.1	57	0.4
5250–5499	127	0.04	31	0.2
5500 or more	59	0.02	14	0.09
Total	354,509		15,501	

*Printed as "4500–4649"; the next row starts at 4750, so the band is 4500–4749.

Detection

- **Macrosomia cannot be diagnosed until delivery** — fundal height and sonography are inaccurate, especially with **obesity**.
- EFW >4000 g within 2 weeks of delivery: **>50%** overestimated; of cesareans done for suspected LGA, **~30%** of babies were **<4000 g**.

Pathophysiology

- **Pedersen hypothesis (1954)**: maternal **hyperglycemia** → **fetal hyperinsulinemia** → **overgrowth**.
- Diabetes + ↑ cord **IGF-1** → more fat mass and cardiac changes (↑ **interventricular septal thickness** with GDM).
- **Persistent pulmonary hypertension of the newborn**: independent risk factors = maternal obesity, diabetes, and both deficient and excessive fetal growth.

Table 47-3. Risk Factors for Fetal Macrosomia

Risk factors
Obesity
Diabetes
Postterm gestation
Multiparity
Large size of parents
Advancing maternal age
Previous macrosomic infant
Racial and ethnic factors

- Many are interrelated (age ↔ parity and diabetes; obesity ↔ diabetes).
- Obese women in China: macrosomia **>24%**; **~2.5×** with prolonged pregnancy and GDM.
- **Diabetes is strongly linked to >4000 g** (Table 47-2), and **type 1 > type 2** for >90th and >97.7th percentiles. Third-trimester glucose, **HbA1c** and **fasting glucose** independently predict it.
- But diabetes accounts for only a **small share** of all LGA newborns.

Management

- **Exercise** ↓ LGA / >4000 g **without** ↑ SGA or <2500 g (28 studies, 5322 women).

- Diabetes: insulin and glycemic control ↓ macrosomia but not consistently cesarean.
- Macrosomia is strongly tied to **maternal obesity** and **excessive gestational weight gain** (Ch 10).

"Prophylactic" labor induction

- Systematic review (11 studies): induction for suspected macrosomia ↑ **cesarean** with no perinatal benefit.
- Metaanalysis (4 RCTs, 1190 women): induction at **≥38 wk** ↓ overgrowth and **fractures**. Boulvain 2015 (822 women): marginally more vaginal births, lower composite morbidity.
- Early-term elective delivery worsens neonatal outcomes → **ACOG: no early induction or delivery before 39 wk for suspected macrosomia**.

Elective cesarean

- **9%** of shoulder-dystocia–related injuries still follow cesarean.
- Nondiabetic women: planned cesarean to prevent **brachial plexopathy** is unreasonable — too many cesareans per injury prevented.
- Diabetic women with EFW **>4250 or >4500 g**: planned cesarean may be reasonable. Conway and Langer: routine cesarean at **≥4250 g** ↓ shoulder dystocia **2.4% → 1.1%**.
- **Elective cesarean is not indicated if EFW <5000 g (no diabetes) or <4500 g (diabetes)** (ACOG).

Outcomes

- Birthweight **≥4000 g** → **cesarean >50%**, especially with obesity, diabetes or >5000 g.
- ↑ **Shoulder dystocia, brachial plexus injury, fractures**; shoulder dystocia up to **~30%** with diabetes; **≥5%** at ≥5000 g in general populations.
- ↑ **Stillbirth** (rising with weight), **PPH, perineal laceration, maternal infection**.

Table 47-4. Maternal and Fetal Outcomes for 208,090 Pregnancies Delivered at Parkland Hospital from 1998 through 2012

Outcome ^a	<4000 g (n = 187,119)	4000–4499 g (n = 17,750)	4500–4999 g (n = 2849)	≥5000 g (n = 372)	p value
Cesarean total	46,577 (25)	5,362 (30)	1204 (42)	224 (60)	<0.001
— Scheduled	12,564 (7)	1,481 (8)	316 (11)	65 (17)	<0.001
— Dystocia	7589 (4)	1388 (8)	337 (12)	46 (12)	<0.001
Shoulder dystocia	437 (0)	366 (2)	192 (7)	56 (15)	<0.001
3rd- or 4th-degree laceration	7296 (4)	932 (5)	190 (7)	37 (10)	<0.001
Labor induction	26,118 (13)	2499 (14)	420 (15)	39 (10)	0.141
Prolonged second stage	6905 (4)	899 (5)	147 (5)	14 (4)	<0.001
Chorioamnionitis	13,448 (7)	1778 (10)	295 (10)	35 (9)	<0.001
pH <7.0	925 (0.5)	96 (0.6)	20 (0.7)	4 (1.1)	0.039
Apgar <7 @ 5 minutes	1898 (1.0)	80 (0.5)	22 (0.8)	10 (2.7)	<0.001
ICN admission	4266 (2.2)	123 (0.7)	36 (1.3)	9 (2.4)	<0.001
Fractured clavicle	1880 (1.0)	616 (3.5)	125 (4.4)	16 (4.3)	<0.001
Mechanical ventilation	2305 (1.2)	54 (0.3)	11 (0.4)	9 (2.4)	<0.001
Hypoglycemia	480 (0.2)	89 (0.5)	31 (1.1)	12 (3.2)	<0.001
Hyperbilirubinemia	5829 (3.0)	305 (1.7)	60 (2.1)	12 (3.2)	<0.001
Erb palsy	470 (0.2)	224 (1.3)	74 (2.6)	22 (5.9)	<0.001
Neonatal death	402 (0.2)	3 (0)	2 (0.1)	1 (0.3)	<0.001

^aOutcome data presented as n (%). ICN = intensive care nursery.

Quick-Recall: FGR vs Macrosomia

Feature	Fetal-growth restriction	Macrosomia
Definition	EFW or AC <10th %ile (prenatal); SGA = birthweight <10th %ile	EFW/birthweight >4000 / 4500 / 5000 g ; LGA = >90th %ile
Truly pathological	Mostly <3rd %ile ; up to 70% of SGA are normal	>2 SD (~3%, ≈4500 g at 40 wk)
Screening	Serial fundal height (sensitivity 11–25%); no routine 3rd-trimester scan	Clinical/sonographic EFW (>50% overestimated)
Key causes	Placental (preeclampsia, APS), smoking, infection (CMV, rubella), aneuploidy (T18 >80%), gastroschisis	Obesity, diabetes , postterm, multiparity, prior macrosomia
Monitoring	Growth q3 wk; weekly UA Doppler + fluid + NST/BPP	Glycemic control, exercise, weight-gain limits
Delivery	Normal Doppler 37–39 wk; AEDF 33–34; REDF 30–32; oligohydramnios 36–37 ^{6/7}	No induction before 39 wk ; cesarean only if EFW ≥5000 g (≥4500 g with diabetes)
Neonatal risks	Hypothermia, hypoglycemia, polycythemia, hyperviscosity, stillbirth	Shoulder dystocia, Erb palsy, clavicle fracture, hypoglycemia, stillbirth
Ineffective therapies	Bed rest, O ₂ , nutrients, volume expansion, heparin, aspirin	Prophylactic early induction, elective cesarean for EFW below thresholds

Key Numbers to Memorize

- Growth phases: hyperplasia **0–16 wk**; + hypertrophy to **32 wk**; growth **5 / 15 / 30 g/d** at 15 / 24 / 34 wk; velocity peaks **35 wk**
- FGR = EFW or AC **<10th %ile**; adverse outcomes mainly **<3rd %ile**; **70%** of SGA normal
- Fundal height lag **≥3 cm** → ultrasound; sensitivity **11–25%**; FGR recurrence **~20%**
- Smoking **–200 g** (2–3× FGR); altitude **–150 g/1000 m**; Dutch famine **–250 g**
- Chronic villitis with C4d **88% vs 5%**; infections **≤5%** of FGR; gastroschisis **1/3** FGR
- SGA in T21 **15–30%**, T13 **50%**, T18 **>80%**
- UA Doppler: adverse outcome **1% vs 12%**; stillbirth AEDF **7%**, REDF **19%**; DV abnormal **20%**, reversed a-wave **46%**
- Deliver: normal Doppler **38–39 wk** (3rd–9th) / **37 wk** (<3rd); AEDF **33–34**; REDF **30–32**; oligohydramnios **36°/7–37°/7**
- Early-onset FGR **<32 wk** (~30%); <1st percentile survival to discharge **14%**
- 90th percentile at 39 wk **~3900 g**; ≥4500 g **1.4%** at Parkland; ≥4000 g → cesarean **>50%**
- Cesarean for macrosomia only if EFW **≥5000 g** (no diabetes) / **≥4500 g** (diabetes); Conway–Langer shoulder dystocia **2.4 → 1.1%**
- ≥5000 g at Parkland: cesarean **60%**, shoulder dystocia **15%**, Erb palsy **5.9%**